

лабораторной работы №1

Установка и конфигурация операционной системы на виртуальную машину

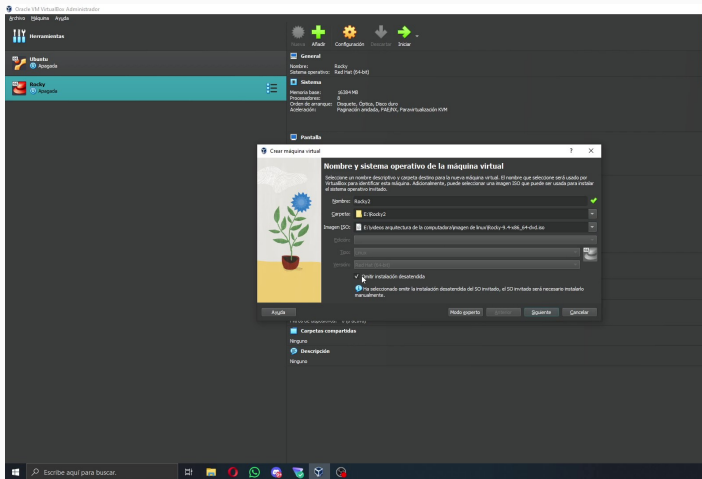
Кхари Жекка Кализая Арсе

29 марта 2025 г.

характеристики ВМ

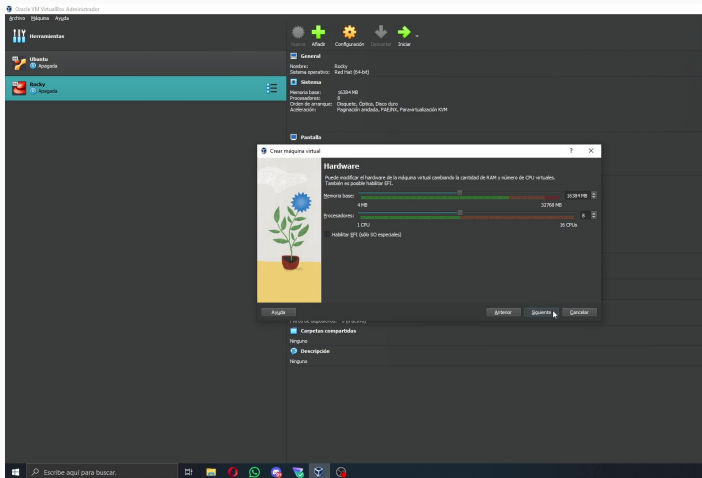
название машины

- название машины
- файл .iso
- расположение машины



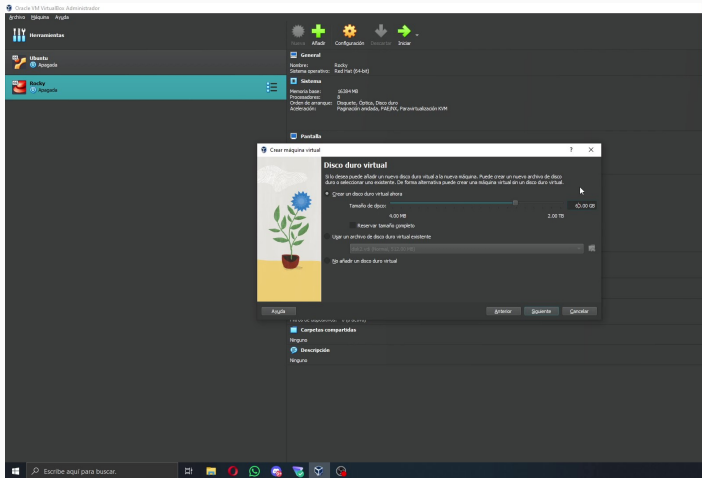
объем памяти и ядра процессора

- 8гб ОП
- 8 ядер процессора



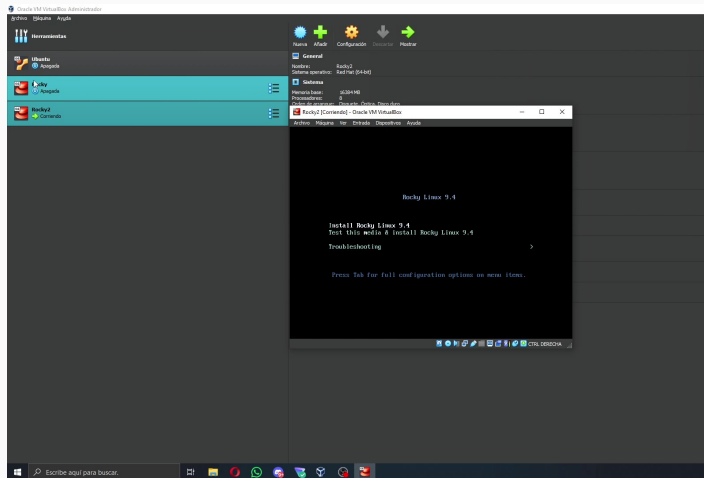
Объем жесткого диска

- 100 гб



Установка ОС

- начинаем настройку ОС



выбор языка

Oracle VM VirtualBox Administrator

Rocky2 [Corriendo] - Oracle VM VirtualBox

Archivo Máquina Ver Entrada Dispositivos Ayuda

Herramientas

Ubuntu Apagada

Rocky Apagada

Rocky2 Corriendo

Rocky Linux

ROCKY LINUX 9.4 INSTALLATION

us Help!

WELCOME TO ROCKY LINUX 9.4.

What language would you like to use during the installation process?

Русский	Russian
العربية	Arabic
English	English
Français	French
Deutsch	German
日本語	Japanese
中文	Mandarin Chinese
Español	Spanish
Afrikaans	Afrikaans
አማርኛ	Amharic
অসমীয়া	Assamese
Asturianu	Asturian
Беларуская	Belarusian
Български	Bulgarian
বাংলা	Bangla
Català	Catalan

Type here to search.

English (United States)

English (United Kingdom)

English (India)

English (Australia)

English (Canada)

English (Denmark)

English (Ireland)

English (New Zealand)

English (Nigeria)

English (Hong Kong SAR China)

English (Philippines)

English (Singapore)

English (South Africa)

English (Zambia)

English (Zimbabwe)

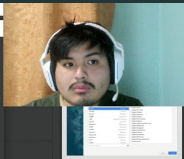
English (Botswana)

English (Antigua & Barbuda)

English (Israel)

Quit Continue

19:20 20/2/2025



выбор жесткого диска

Oracle VM VirtualBox Administrator

Rocky2 [Corriendo] - Oracle VM VirtualBox

Archivo Máquina Ver Entrada Dispositivos Ayuda

INSTALLATION DESTINATION

Done Help!

us

Device Selection

Select the device(s) you'd like to install to. They will be left untouched until you click on the main menu's "Begin Installation" button.

Local Standard Disks

60 GiB

ATA VBOX HARDISK
sda / 60 GiB free

Disks left unselected here will not be touched.

Specialized & Network Disks

Add a disk...

Disks left unselected here will not be touched.

Storage Configuration

☒ Automatic ☐ Custom

☐ I would like to make additional space available.

Encryption

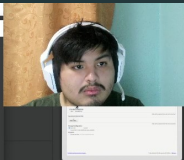
☐ Encrypt my data. You'll set a passphrase next.

[Full disk summary and boot loader...](#)

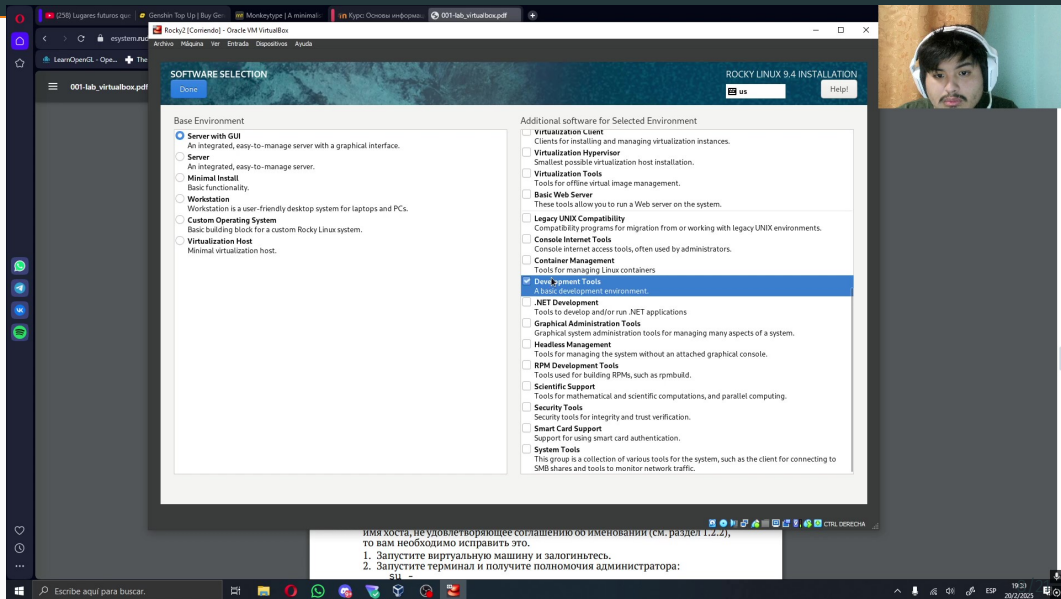
1 disk selected; 60 GiB capacity; 60 GiB free [Refresh...](#)

CTRL DERECHA

19:29 20/2/2025



установка программного обеспечения



Rocky Linux 9.4 INSTALLATION

us Help!

SOFTWARE SELECTION

Done

Base Environment

- ☒ **Server with GUI**
An integrated, easy-to-manage server with a graphical interface.
- ☐ **Server**
An integrated, easy-to-manage server.
- ☐ **Minimal Install**
Basic functionality.
- ☐ **Workstation**
Workstation is a user-friendly desktop system for laptops and PCs.
- ☐ **Custom Operating System**
Basic building block for a custom Rocky Linux system.
- ☐ **Virtualization Host**
Minimal virtualization host.

Additional software for Selected Environment

- ☐ **Virtualization Client**
Clients for installing and managing virtualization instances.
- ☐ **Virtualization Hypervisor**
Smallest possible virtualization host installation.
- ☐ **Virtualization Tools**
Tools for offline virtual image management.
- ☐ **Basic Web Server**
These tools allow you to run a Web server on the system.
- ☐ **Legacy UNIX Compatibility**
Compatibility programs for migration from or working with legacy UNIX environments.
- ☐ **Console Internet Tools**
Console internet access tools, often used by administrators.
- ☐ **Container Management**
Tools for managing Linux containers.
- ☒ **Development Tools**
A basic development environment.
- ☐ **.NET Development**
Tools to develop and/or run .NET applications.
- ☐ **Graphical Administration Tools**
Graphical system administration tools for managing many aspects of a system.
- ☐ **Headless Management**
Tools for managing the system without an attached graphical console.
- ☐ **RPM Development Tools**
Tools used for building RPMs, such as rpmbuild.
- ☐ **Scientific Support**
Tools for mathematical and scientific computations, and parallel computing.
- ☐ **Security Tools**
Security tools for integrity and trust verification.
- ☐ **Smart Card Support**
Support for using smart card authentication.
- ☐ **System Tools**
This group is a collection of various tools for the system, such as the client for connecting to SMB shares and tools to monitor network traffic.

имя хоста, не удовлетворяющее соглашению об именовании (см. раздел 1.2.2),
то вам необходимо исправить это.

1. Запустите виртуальную машину и залогиньтесь.
2. Запустите терминал и получите полномочия администратора:

su -

отключение KDUMP

The screenshot shows a virtual machine window titled "Rocky2 [Corriendo] - Oracle VM VirtualBox". Inside the VM, the "KDUMP" configuration screen for Rocky Linux 9.4 is displayed. The text explains that kdump is a kernel crash dumping mechanism and that enabling it requires reserving system memory. The "Enable kdump" checkbox is currently unchecked. The window has a sidebar with navigation links like "Archivo", "Máquina", "Ver", "Entrada", "Dispositivos", and "Ayuda". At the bottom of the VM window, there is a "Specialized & Network Disks" section with an "Add a disk..." button. The overall image is a video call frame, with a person wearing headphones visible in the top right corner. The Windows taskbar at the bottom shows the search bar with the text "Escribe aquí para buscar.", several application icons, and the system clock indicating 19:30 on 20/2/2025.

изменение названия хоста

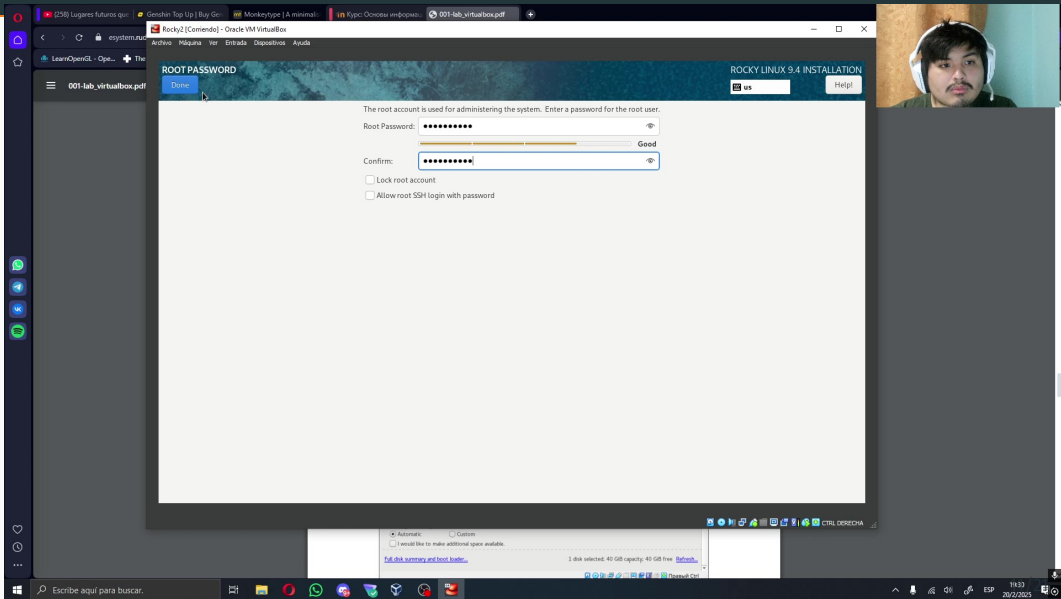
The screenshot shows a Rocky Linux 9.4 installation window titled "Rocky2 [Corriendo] - Oracle VM VirtualBox". The window is divided into two main sections: "NETWORK & HOST NAME" and "ROCKY LINUX 9.4 INSTALLATION".

In the "NETWORK & HOST NAME" section, the "Ethernet (enp0s3)" interface is selected. The "Host Name" field is set to "qjarisheikka". The "Current host name" is also "qjarisheikka". The "Apply" button is highlighted.

The "ROCKY LINUX 9.4 INSTALLATION" section shows the "us" locale selected and a "Help!" button.

The bottom of the window shows the system tray with various icons and the date "20/2/2025".

настройка пароля



The screenshot shows the Rocky Linux 9.4 installation window. The title bar reads "Rocky2 [Corriendo] - Oracle VM VirtualBox". The window title is "ROOT PASSWORD". In the top right corner, it says "ROCKY LINUX 9.4 INSTALLATION" with a language dropdown set to "us" and a "Help!" button. The main text says: "The root account is used for administering the system. Enter a password for the root user."

There are two password input fields:

- "Root Password:" with a strength indicator showing "Good".
- "Confirm:"

Below the fields are two checkboxes:

- ☐ Lock root account
- ☐ Allow root SSH login with password

At the bottom of the window, there is a disk selection summary:

- 1 disk selected: 40 GB capacity: 40 GB free
- Buttons: [Full disk summary and boot loader...](#) and [Refresh...](#)

The bottom of the screen shows the Windows taskbar with the search bar "Escribe aquí para buscar." and the system clock showing "11:33 20/2/2025".

настройка пользователя

The screenshot shows the 'CREATE USER' screen during the Rocky Linux 9.4 installation. The window title is 'Rocky2 [Corriendo] - Oracle VM VirtualBox'. The browser address bar shows 'system.zuc'. The page has a dark blue header with 'CREATE USER' and 'ROCKY LINUX 9.4 INSTALLATION'. A 'Done' button is in the top left, and a 'Help!' button is in the top right. The form contains the following fields and options:

- Full name: qjarishekkka
- User name: qjarishekkka
- ☒ Make this user administrator
- ☒ Require a password to use this account
- Password: [masked with dots] (Strength: Good)
- Confirm password: [masked with dots]
- Advanced... button

At the bottom of the window, there is a summary section with the following text:

☒ Automatic ☐ Custom
☐ I would like to make additional space available.
[Full disk summary and boot loader...](#)
1 disk selected: 40 GB capacity: 40 GB free [Refresh...](#)

In the top right corner, there is a video call window showing a person with dark hair and a beard wearing large white headphones.

начало установки

The screenshot shows the Rocky Linux 9.4 installation summary screen within a virtual machine. The window title is "Rocky2 [Corriendo] - Oracle VM VirtualBox". The browser address bar shows "system.zuc". The left sidebar contains a file manager with "001-lab_virtualbox.pdf" and a social media sidebar with icons for WhatsApp, Telegram, and others. The main content area displays the "INSTALLATION SUMMARY" for "ROCKY LINUX 9.4 INSTALLATION".

INSTALLATION SUMMARY

us Help!

LOCALIZATION

- Keyboard: English (US)
- Language Support: English (United States)
- Time & Date: Europe/Moscow timezone

SOFTWARE

- Installation Source: Local media
- Software Selection: Server with GUI

SYSTEM

- Installation Destination: Automatic partitioning selected
- KDUMP: Kdump is disabled
- Network & Host Name: Connected - mp0k3
- Security Profile: No profile selected

USER SETTINGS

- Root Password: Root password is set
- User Creation: Administrator gjenishehka will be created

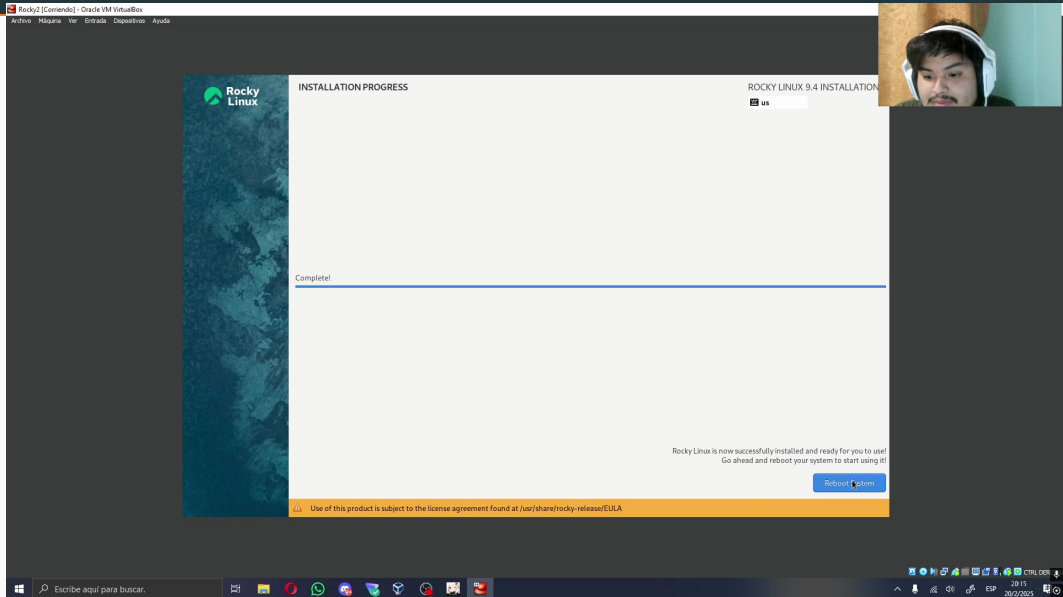
Out Begin Installation

We won't touch your disks until you click 'Begin Installation'

At the bottom, a terminal window shows the command `[user@user ~]$ sudo -i` and the output `We trust you have received the usual lecture from the local System Administrator. It usually boils down to these three things:`

The system tray at the bottom shows the Windows taskbar with the search bar "Escribe aquí para buscar.", taskbar icons, and the system clock showing 19:23 on 20/2/2025.

перезагрузка системы



терминал, VMB и основная
информация системы

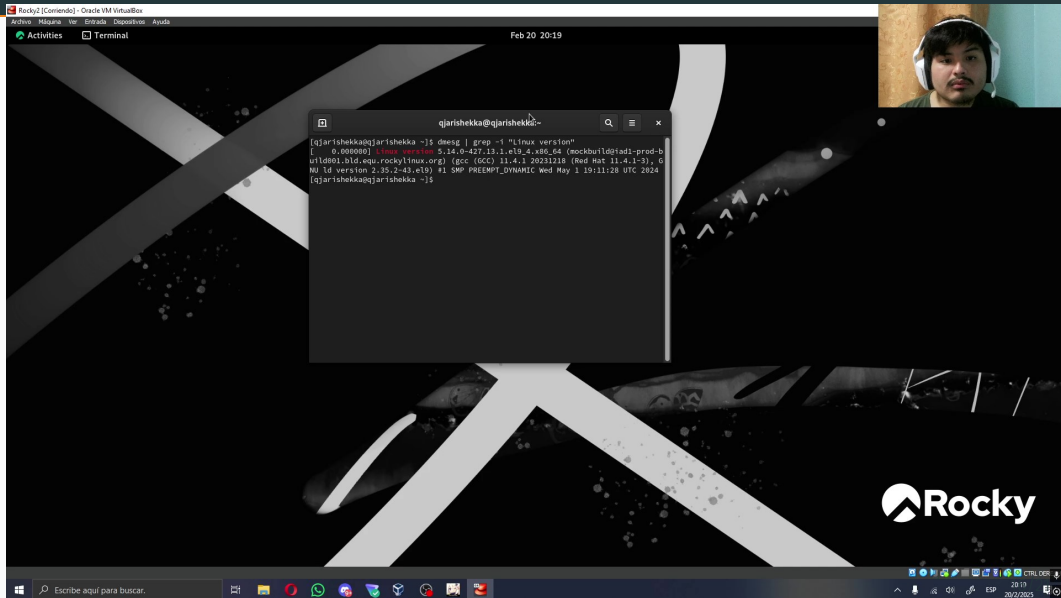
установка плагинов VMB

Rocky2 [Comiendo] - Oracle VM VirtualBox
Archivo Máquina Ver Entrada Dispositivos Ayuda



```
Activities Terminal Feb 20 20:18
VirtualBox Guest Additions installation
Verifying archive integrity... 100% MD5 checksums are OK. All good.
Uncompressing VirtualBox 7.0.18 Guest Additions for Linux 100%
VirtualBox Guest Additions installer
Copying additional installer modules ...
Installing additional modules ...
VirtualBox Guest Additions: Starting.
VirtualBox Guest Additions: Setting up modules
VirtualBox Guest Additions: Building the VirtualBox Guest Additions kernel
modules. This may take a while.
VirtualBox Guest Additions: To build modules for other installed kernels, run
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup <version>
VirtualBox Guest Additions: or
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup all
VirtualBox Guest Additions: Building the modules for kernel
5.14.0-427.13.1.el9_4.x86_64
VirtualBox Guest Additions: reloading kernel modules and services
VirtualBox Guest Additions: kernel modules and services 7.0.18 r162988 reloaded
VirtualBox Guest Additions: NOTE: you may still consider to re-login if some
user session specific services (Shared Clipboard, Drag and Drop, Seamless or
Guest Screen Resize) were not restarted automatically
Press Return to close this window...
```

версия linux



The image shows a screenshot of a Rocky Linux virtual machine running in Oracle VM VirtualBox. The desktop background features a large, stylized 'X' pattern. A terminal window is open in the center, displaying the output of the command `dmesg | grep -i "Linux version"`. The output shows the kernel version `5.14.0-427.13.1.el9_4.x86_64` and other system details. In the top right corner, there is a small video feed of a person wearing headphones. The bottom of the screen shows the Windows taskbar with various application icons and the system clock.

Rocky2 [Comiendo] - Oracle VM VirtualBox

Archivo Máquina Ver Entrada Dispositivos Ayuda

Activities Terminal

Feb 20 20:19

qjarishekk@qjarishekkai-~

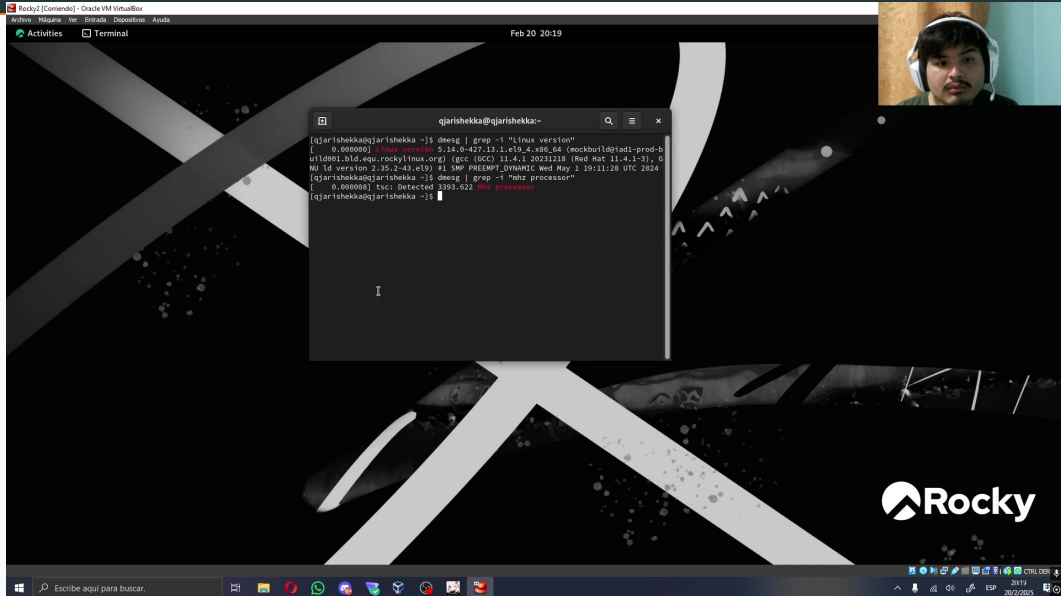
```
[qjarishekk@qjarishekkai ~]$ dmesg | grep -i "Linux version"
[ 0.000000] Linux version 5.14.0-427.13.1.el9_4.x86_64 (mockbuild@iad1-prod-b
uid801.bld.equ.rockylinux.org) (gcc (GCC) 11.4.1 20231218 (Red Hat 11.4.1-3), G
NU ld version 2.35.2-43.el9) #1 SMP PREEMPT_DYNAMIC Wed May 1 19:11:28 UTC 2024
[qjarishekk@qjarishekkai ~]$
```

Rocky

Escribe aquí para buscar.

20:19 20/2/2025

MHZ процессора



The image shows a screenshot of a Rocky Linux virtual machine running in Oracle VM VirtualBox. The VM's desktop background features a large, stylized 'X' pattern. A terminal window is open in the center, displaying the output of two commands. The first command, `dmesg | grep -i "Linux version"`, shows the kernel version and build information. The second command, `dmesg | grep -i "mhz processor"`, shows the detected processor frequency. In the top right corner, there is a small video feed of a person wearing headphones. The bottom of the screen shows the Windows taskbar with various application icons and the system clock.

Rocky2 [Comiendo] - Oracle VM VirtualBox

Archivo Máquina Ver Entrada Dispositivos Ayuda

Activities Terminal Feb 20 20:19

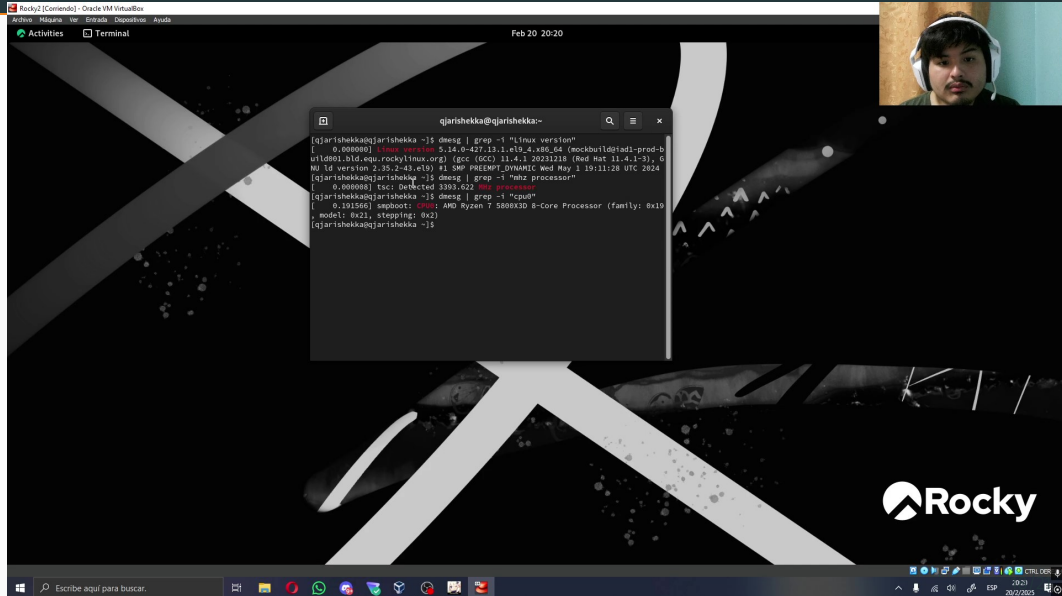
```
qjarishekk@qjarishekk:~  
[qjarishekk@qjarishekk ~]$ dmesg | grep -i "Linux version"  
[ 0.000000] Linux version 5.14.0-427.13.1.el9_4.x86_64 (mockbuild@iad1-prod-b  
uild001.bld.equ.rockylinux.org) (gcc (GCC) 11.4.1 20231218 (Red Hat 11.4.1-3), G  
NU ld version 2.35.2-43.el9) #1 SMP PREEMPT_DYNAMIC Wed May 1 19:11:28 UTC 2024  
[qjarishekk@qjarishekk ~]$ dmesg | grep -i "mhz processor"  
[ 0.000000] tsc: Detected 3393.622 Mhz processor  
[qjarishekk@qjarishekk ~]$
```

Rocky

Escribe aquí para buscar.

20/2/2025

тип процессора



The image shows a screenshot of a Rocky Linux virtual machine running in Oracle VM VirtualBox. The desktop background features a large, stylized 'X' pattern. A terminal window is open in the center, displaying the output of several system information commands. The terminal output includes the Linux version (5.14.0-427.13.1.el9_4.x86_64), the tsc (3393.622 MHz), and the CPU information (AMD Ryzen 7 5800X3D 8-Core Processor). The Rocky Linux logo is visible in the bottom right corner of the desktop. A small video feed of a person wearing headphones is visible in the top right corner of the screen.

Rocky2 [Comiendo] - Oracle VM VirtualBox

Archivo Máquina Ver Entrada Dispositivos Ayuda

Activities Terminal

Feb 20 20:20

qjarishekk@qjarishekk:~

```
[qjarishekk@qjarishekk ~]$ dmesg | grep -i "Linux version"
[    0.000000] Linux version 5.14.0-427.13.1.el9_4.x86_64 (mockbuild@iad1-prod-b
wild001.bld.equ.rockylinux.org) (gcc (GCC) 11.4.1 20231218 (Red Hat 11.4.1-3), G
NU ld version 2.35.2-43.el9) #1 SMP PREEMPT_DYNAMIC Wed May 1 19:11:28 UTC 2024
[qjarishekk@qjarishekk ~]$ dmesg | grep -i "mhz processor"
[    0.000000] tsc: Detected 3393.622 MHz processor
[qjarishekk@qjarishekk ~]$ dmesg | grep -i "cpu0"
[    0.191566] smpboot: CPU0: AMD Ryzen 7 5800X3D 8-Core Processor (family: 0x19
, model: 0x21, stepping: 0x2)
[qjarishekk@qjarishekk ~]$
```

Rocky

Escribe aquí para buscar.

20:21 20/2/2025

свободная оперативная память

Rocky2 [Comiendo] - Oracle VM VirtualBox

Archivo Máquina, Ver, Entrada, Dispositivos, Ayuda

Activities Terminal

Feb 20 20:20

qjarishekk@qjarishekk:~

```
[ 0.013773] PM: hibernation: Registered nosave memory: [mem 0xfec0b080-0xfec0bfff]
[ 0.013773] PM: hibernation: Registered nosave memory: [mem 0xfec01080-0xfedffff]
[ 0.013774] PM: hibernation: Registered nosave memory: [mem 0xfec0b080-0xfec0bfff]
[ 0.013774] PM: hibernation: Registered nosave memory: [mem 0xfec01080-0xfedffff]
[ 0.013774] PM: hibernation: Registered nosave memory: [mem 0xffff0080-0xfffffff]
[ 0.044880] Memory: 3677188K/16776760K available (16384K kernel code, 5626K r
wdata, 11748K rodata, 3892K init, 5956K bss, 464396K reserved, 0K cma-reserved)
[ 0.087487] Freeing SMP alternatives memory: 36K
[ 0.222483] x86/mm: Memory block size: 128MB
[ 0.361653] Non-volatile memory driver v1.3
[ 0.731716] Freeing initrd memory: 57896K
[ 0.884411] Freeing unused decrypted memory: 2028K
[ 0.884978] Freeing unused kernel image (initmem) memory: 3892K
[ 0.908048] Freeing unused kernel image (rodata/data gap) memory: 548K
[ 2.215122] vmwgfx 0800:00:02.0: [drm] Legacy memory limits: VRAM = 16384 kB,
FIFO = 2048 kB, surface = 507904 kB
[ 2.215126] vmwgfx 0800:00:02.0: [drm] Maximum display memory size is 16384 k
B
[qjarishekk@qjarishekk ~]$
```

Rocky

Escribe aquí para buscar.

20:20
20/2/2025

Hypervisor

Rocky2 [Comiendo] - Oracle VM VirtualBox

Archivo Máquina Ver Entrada Depósitos Ayuda

Activities Terminal

Feb 20 20:21

qjarishekkka@qjarishekkka:~

```
[qjarishekkka@qjarishekkka ~]$ dmesg | grep -i "memory available"
[qjarishekkka@qjarishekkka ~]$ dmesg | grep -i "memory"
[ 0.001647] ACPI: Reserving FACP table memory at [mem 0xdffff08f8-0xdffff1e3]
[ 0.001648] ACPI: Reserving DSDT table memory at [mem 0xdffff0640-0xdffff2992]
[ 0.001648] ACPI: Reserving FACS table memory at [mem 0xdffff0200-0xdffff023f]
[ 0.001649] ACPI: Reserving FACS table memory at [mem 0xdffff0200-0xdffff023f]
[ 0.001649] ACPI: Reserving APIC table memory at [mem 0xdffff0240-0xdffff02cb]
[ 0.001650] ACPI: Reserving SSDT table memory at [mem 0xdffff02d0-0xdffff063b]
[ 0.001924] Early memory node ranges
[ 0.013769] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
[ 0.013771] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
[ 0.013771] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
[ 0.013772] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
[ 0.013772] PM: hibernation: Registered nosave memory: [mem 0xdffff0000-0xdffff0fff]
[ 0.013773] PM: hibernation: Registered nosave memory: [mem 0xe0000000-0xfefbffff]
[ 0.013773] PM: hibernation: Registered nosave memory: [mem 0xfec00000-0xfec00fff]
[ 0.013774] PM: hibernation: Registered nosave memory: [mem 0xfec00000-0xfec00fff]
[ 0.013774] PM: hibernation: Registered nosave memory: [mem 0xfec00000-0xfec00fff]
[ 0.013774] PM: hibernation: Registered nosave memory: [mem 0xfec00000-0xfec00fff]
[ 0.013774] PM: hibernation: Registered nosave memory: [mem 0xfec00000-0xfec00fff]
[ 0.044801] Memory: 367718K/1677676K available (16384K kernel code, 5626K rdata, 11748K rodata, 3892K init, 5956K bss, 464396K reserved, 0K cma-reserved)
[ 0.087487] Freeing SMP alternatives memory: 36K
[ 0.222483] x86/mm: Memory block size: 128MB
[ 0.361653] Non-volatile memory driver v1.3
[ 0.731716] Freeing initrd memory: 57096K
[ 0.884411] Freeing unused decrypted memory: 2028K
[ 0.884978] Freeing unused kernel image (initmem) memory: 3892K
[ 0.908048] Freeing unused kernel image (rodata/data gap) memory: 548K
[ 2.215122] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 16384 kB, FIFO = 2048 kB, surface = 507904 kB
[ 2.215126] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 16384 KiB
[qjarishekkka@qjarishekkka ~]$ dmesg | grep -i "hypervisor detected"
[ 0.000000] Hypervisor detected: KVM
[qjarishekkka@qjarishekkka ~]$
```

Rocky

Escribe aquí para buscar.

20:21 20/2/2025

filesystem

Rocky2 [Comiendo] - Oracle VM VirtualBox

Archivo Máquina Ver Entrada Depósitos Ayuda

Activities Terminal

Feb 20 20:21

qjarishekk@qjarishekk:~

```
[ 0.001648] ACPI: Reserving FACS table memory at [mem 0xdffff0200-0xdffff023f]
[ 0.001649] ACPI: Reserving FACS table memory at [mem 0xdffff0200-0xdffff023f]
[ 0.001649] ACPI: Reserving APIC table memory at [mem 0xdffff0240-0xdffff02cb]
[ 0.001650] ACPI: Reserving SSDT table memory at [mem 0xdffff02d0-0xdffff063b]
[ 0.001924] Early memory node ranges
[ 0.013769] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
[ 0.013771] PM: hibernation: Registered nosave memory: [mem 0x0009f000-0x0009ffff]
[ 0.013771] PM: hibernation: Registered nosave memory: [mem 0x000a0000-0x000aefff]
[ 0.013772] PM: hibernation: Registered nosave memory: [mem 0x000f0000-0x000fffff]
[ 0.013772] PM: hibernation: Registered nosave memory: [mem 0xdfff0000-0xdfffffff]
[ 0.013773] PM: hibernation: Registered nosave memory: [mem 0xe0000000-0xfefbffff]
[ 0.013773] PM: hibernation: Registered nosave memory: [mem 0xfec00000-0xfec0ffff]
[ 0.013773] PM: hibernation: Registered nosave memory: [mem 0xfec01000-0xfedfffff]
[ 0.013774] PM: hibernation: Registered nosave memory: [mem 0xfec00000-0xfec0ffff]
[ 0.013774] PM: hibernation: Registered nosave memory: [mem 0xfec01000-0xfefbffff]
[ 0.013774] PM: hibernation: Registered nosave memory: [mem 0xfec00000-0xfec0ffff]
[ 0.044881] Memory: 867108K/1677660K available (16384K kernel code, 5620K rwdata, 11748K rodata, 3892K init, 5956K b
ss, 464396K reserved, 0K cma-reserved)
[ 0.087487] Freeing SMP alternatives memory: 36K
[ 0.222483] x86/mm: Memory block size: 128MB
[ 0.361653] Non-volatile memory driver v1.3
[ 0.731716] Freeing initrd memory: 57096K
[ 0.884411] Freeing unused decrypted memory: 2028K
[ 0.884978] Freeing unused kernel image (initmem) memory: 3892K
[ 0.908048] Freeing unused kernel image (rodata/data gap) memory: 548K
[ 2.215122] vmx: 0000:00:02.0: [drm] legacy memory limits: VRAM = 16384 kB, FIFO = 2048 kB, surface = 507904 kB
[ 2.215126] vmx: 0000:00:02.0: [drm] Maximum display memory size is 16384 kib
qjarishekk@qjarishekk ~$ dmesg | grep -i "hypervisor detected"
[ 0.000000] Hypervisor detected: KVM
qjarishekk@qjarishekk ~$ dmesg | grep -i "filesystem"
[ 4.279068] XFS (dm-0): Mounting V5 Filesystem 353e9c3e-f6e8-48f7-bbe2-6f0c6d29eaa4
[ 6.658333] XFS (sdal): Mounting V5 Filesystem 474719c7-bac5-49aa-81cd-35ab029f9bf8
[ 6.658588] XFS (dm-2): Mounting V5 Filesystem 7e3d236d-f0a1-4030-bb78-289f7dcd0256
qjarishekk@qjarishekk ~$
```

Rocky

Escribe aquí para buscar.

26/21
20/2/2025